



Student number: _____

GIRRAWEEEN HIGH SCHOOL

2021 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION,

MATHEMATICS EXTENSION 1

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- NESA approved calculators may be used.
- A reference sheet is provided at the back of this paper.
- In section II, Show relevant mathematical reasoning and/or calculations

Total marks: 70

Section I – 10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 60 marks

- Attempt all questions
- Allow about 1 hour and 45 minutes for this section

SECTION 1**10 marks****Attempt questions 1 – 10****Allow about 15 minutes for this section****Use the multiple-choice answer sheet for questions 1 - 10**

1. Given that $\underline{x} = 5\underline{i} + 3\underline{j}$ and $\underline{y} = -2\underline{i} - 5\underline{j}$. The magnitude and direction of $\underline{x} + \underline{y}$ is

(A) 3.6; 326° **(B)** 3.6; 34° **(C)** 3.6; 146° **(D)** 3.6; 214°

2. In the expansion of $(2x + k)^6$, the coefficients of x and x^2 are equal. What is the value of k ?

(A) 5**(B)** 6**(C)** 11**(D)** 12

3. The coefficient of x^{-5} in the expansion of $\left(2x^2 - \frac{1}{x}\right)^{20}$ is

(A) -77520**(B)** -155040**(C)** -248064**(D)** -496128

4. The domain and inverse of $f(x) = 4 \log_e(x+3) - 2$ are

(A) $x > 3; y = e^{\frac{x+2}{4}} - 3$

(B) $x > -3; y = e^{\frac{x+2}{4}} - 2$

(C) $x > -3; y = e^{\frac{x+2}{4}} - 3$

(D) $x > 3; y = e^{\frac{x+2}{4}} - 2$

5. Consider the parametric equation $x = 5 \cos \theta - 2$ and $y = 5 \sin \theta + 3$. Which of these is the corresponding cartesian equation?

(A) $x^2 - 4x + y^2 - 6y = 12$

(B) $x^2 + 4x + y^2 + 6y = 12$

(C) $x^2 - 4x + y^2 + 6y = 12$

(D) $x^2 + 4x + y^2 - 6y = 12$

6. What is the derivative of $y = \cos^{-1}\left(\frac{x}{4}\right)$

(A) $-\frac{1}{\sqrt{16-x^2}}$

(B) $-\frac{2}{\sqrt{16-x^2}}$

(C) $-\frac{4}{\sqrt{16-x^2}}$

(D) $-\frac{6}{\sqrt{16-x^2}}$

7. What is the domain and range of $f(x) = 2 \sin^{-1}\left(\frac{x}{2}\right)$?

(A) $D: -2 \leq x \leq 2, R: -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

(B) $D: -2 \leq x \leq 2, R: -\pi \leq y \leq \pi$

(C) $D: -\frac{1}{2} \leq x \leq \frac{1}{2}, R: -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

(D) $D: -\frac{1}{2} \leq x \leq \frac{1}{2}, R: -\pi \leq y \leq \pi$

8. $\int \sin^2 3x \, dx$ is equal to which of the following?

(A) $\frac{x}{2} - \frac{\sin 6x}{3} + C$

(B) $\frac{x}{2} - \frac{\sin 6x}{6} + C$

(C) $\frac{x}{2} - \frac{\sin 6x}{9} + C$

(D) $\frac{x}{2} - \frac{\sin 6x}{12} + C$

9. What is the value of k such that $\int_0^k \frac{dx}{1+(x-1)^2} = \frac{\pi}{2}$

(A) $2\sqrt{3}$

(B) $\sqrt{3}$

(C) 2

(D) 1

10. Which of the following is a factor of $2x^4 - 4x^3 - 10x^2 + 12x$?

(A) $x+1$

(B) $x-2$

(C) $x-3$

(D) $x+4$

Section II**60 marks****Attempt all questions****Allow about 1 hour and 45 minutes for this section**

Start each question on a new page in the answer booklet provided.

Your responses should include relevant mathematical reasoning and /or calculations. Extra writing space is available on request.

Question 11 (12 marks)**Marks**

(a) Solve $\frac{6}{5x-2} \leq 2$ 3

(b) Prove that $\cot 2x + \cot x = \frac{\sin 3x}{\sin 2x \sin x}$ 2

✓ (c) Use the substitution $u = \ln 3x$, to find $\int \frac{dx}{x(\ln 3x)^2}$ 3

✓ (d) Let $f(x) = \frac{2x}{\sqrt{1-x^2}}$

(i) For what values of x is $f(x)$ undefined? 1

(ii) Find $\int_0^{\frac{1}{2}} \frac{2x dx}{\sqrt{1-x^2}}$ using the substitution $x = \sin u$. 3

End of Question 11

Question 12 (12 marks)

(a) (i) Express $5 \sin x + 12 \cos x$ in the form $A \sin(x + \alpha)$ where $0 \leq \alpha \leq \frac{\pi}{2}$ (Give the value of α in radians, correct to 2 decimal places) 3

(ii) Hence solve $5 \sin x + 12 \cos x = 8$ for $0 \leq x \leq \pi$ (Give the value or values of x in radians correct to 2 decimal places) 2

(b) Six people attend a dinner party.

(i) In how many different ways can they be arranged around a round table? 1

(ii) In how many different ways can they be arranged if a particular couple must sit together? 1

(iii) What is the probability that, if the people are seated at random, the couple are sitting apart from each other? 1

(c) Use mathematical induction to prove that

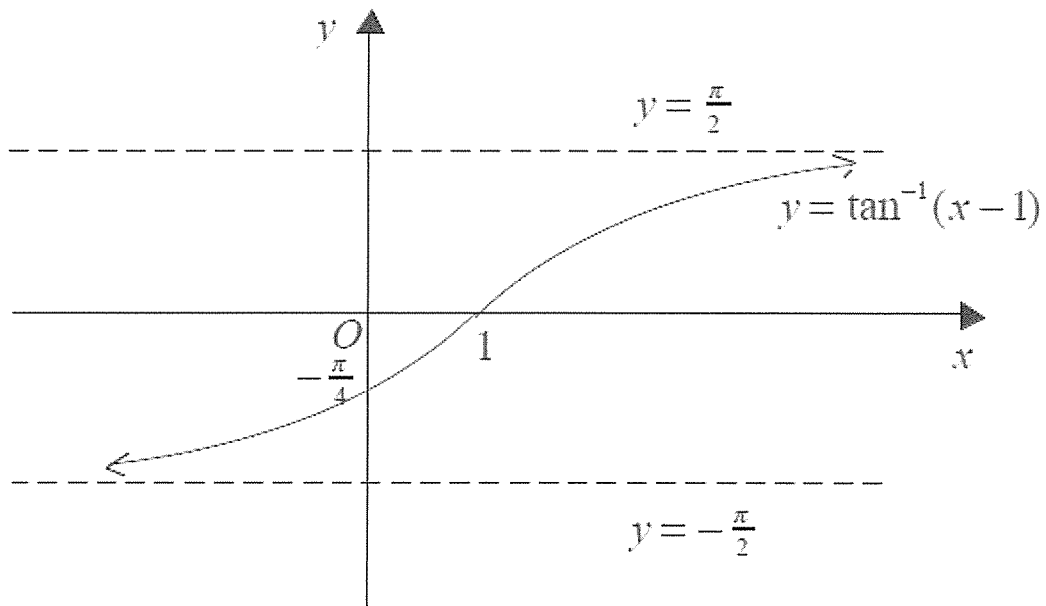
$$(1^2 + 1) 1! + (2^2 + 1) 2! + (3^2 + 1) 3! + \dots + (n^2 + 1) n! = n(n+1)! \text{ for all positive integers}$$

$n \geq 1$. 4

End of Question12

Question 13 (12 marks)

(a)



The region in the first quadrant bounded by the curve $y = \tan^{-1}(x-1)$ and the y -axis between the lines $y = 0$ and $y = \frac{\pi}{4}$ is rotated through one complete revolution about the y -axis.

(i) Show that the volume V of the solid of revolution is given by

$$V = \pi \int_0^{\frac{\pi}{4}} (1 + \tan y)^2 dy. \quad 1$$

(ii) Hence find the value of V in simplest exact form. 3

(b) A particle is projected from a point O with velocity V m/s at an angle θ to the horizontal. At any time t seconds the horizontal and vertical components of displacement are given by

$$x = Vt \cos \theta \text{ and } y = Vt \sin \theta - \frac{1}{2}gt^2 \text{ where } g \text{ is the acceleration due to gravity.}$$

Show that the cartesian equation of the path is given by $y = x \tan \theta - \frac{gx^2}{2V^2}(1 + \tan^2 \theta)$ 2

(c) A particle is projected from O with velocity 60 m/s at an angle α to the horizontal.

T seconds later, another particle is projected from O with velocity 60 m/s at an angle β

To the horizontal where $\beta < \alpha$. The two particles collide 240 metres horizontally from O

and at a height of 80 metres above O . Taking $g = 10 \text{ m/s}^2$ and using results from (a)

(i) Show that $\tan \alpha = 2$ and $\tan \beta = 1$. 3

(ii) Find the value of T in simplest exact form. 3

End of Question 13

Question 14 (12 marks)

(a) (i) Differentiate $y = x \cos^{-1} x - \sqrt{1-x^2}$. 2

(ii) Hence calculate the exact value of $\int_0^{\frac{1}{2}} \cos^{-1} x dx$ 2

(b) Solve $x^4 - 5x^3 - 9x^2 + 81x - 108 = 0$, given that $P(x) = x^4 - 5x^3 - 9x^2 + 81x - 108$

has a triple zero. 3

(c) A bottle of medicine which is initially at a temperature of 10°C is placed into a room which has a constant temperature of 25°C . The medicine warms at a rate proportional to the difference between the temperature of the room and the temperature (T) of the medicine. That is, T satisfies the equation $\frac{dT}{dt} = -k(T - 25)$

(i) Show that $T = 25 + Ae^{-kt}$ is a solution of this equation. 2

(ii) If the temperature of the medicine after 10 minutes is 16°C , find its temperature after 40 minutes. 3

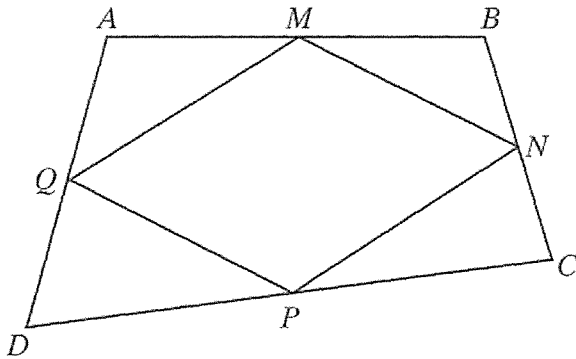
End of Question 14

Question 15 (12 marks)

(a) For what value(s) of m are the vectors $\begin{pmatrix} 10m-17 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} m \\ 2 \end{pmatrix}$ perpendicular? 3

(b) Consider the vectors given by $\underline{u} = b\underline{i} + 2\underline{j}$ and $\underline{w} = 2\underline{i} + b\underline{j}$ where b is a real number.
If the acute angle between the two vectors is 60° , find the two possible values
for b . 4

(c) Consider the quadrilateral $ABCD$. The midpoints of AB, BC, CD and DA are M, N, P and Q respectively.



Let $\overrightarrow{AB} = \underline{a}$, $\overrightarrow{BC} = \underline{b}$, $\overrightarrow{CD} = \underline{c}$ and $\overrightarrow{DA} = \underline{d}$

(i) Prove that $\underline{a} + \underline{b} + \underline{c} + \underline{d} = 0$ 2

(ii) Hence prove that $MNPQ$ is a parallelogram. 3

END OF TEST

Year 12 Trial HSC Extensions 1, 2021 Solutions

1. $x = 5i + 3j$

$y = -2i - 5j$

$x + y = 3i - 2j$

$|x + y| = \sqrt{9 + 4}$
 $= \sqrt{13} = 3.6$

$\tan \theta = \frac{-2}{3}$

acute $\angle = \tan^{-1}(\frac{2}{3})$
 $= 34^\circ$

$\angle = 360 - 34$
 $= 326$ (A)

2. $6C_4 \times 2^2 \times k^4 = 6C_5 \times 2 \times k^5$

$k = \frac{6C_4 \times 2}{6C_5} = 5$ (A)

3. $T_{r+1} = nC_r a^{n-r} b^r$

$(-1)^r {}^{20}C_r (2x^2)^{20-r} (\frac{1}{2x})^r$

$(-1)^r {}^{20}C_r 2^{20-r} x^{40-3r}$

$40 - 3r = -5$

$r = 15$

$(-1)^{15} {}^{20}C_{15} 2^5$

$= -496128$ (D)

4. $x + 3 > 0$

$x > -3$

$y = 4 \log_e (x+3) - 2$

$x = 4 \log_e (y+3) - 2$

$x + 2 = 4 \log_e (y+3)$

$\frac{x+2}{4} = \log_e (y+3)$

$y+3 = e^{\frac{x+2}{4}}$

$y = e^{\frac{x+2}{4}} - 3$ (C)

5. $x = 5 \cos \theta - 2$, $y = 5 \sin \theta + 3$

$\cos \theta = \frac{x+2}{5}$

$\sin \theta = \frac{y-3}{5}$

$\frac{(x+2)^2}{25} + \frac{(y-3)^2}{25} = 1$

$(x+2)^2 + (y-3)^2 = 25$

$x^2 + 4x + 4 + y^2 - 6y + 9 = 25$

$x^2 + 4x + y^2 - 6y = 12$ (D)

6. $y = \cos^{-1}(\frac{x}{4})$

$y' = \frac{-1}{\sqrt{16-x^2}}$ (A)

7. $f(x) = 2 \sin^{-1}(\frac{x}{2})$

D: $-2 \leq x \leq 2$

R: $-\pi \leq y \leq \pi$ (B)

$$8. \int \sin^2 3x \, dx$$

$$= \int \frac{1 - \cos 6x}{2} \, dx$$

$$= \frac{1}{2} \int (1 - \cos 6x) \, dx$$

$$= \frac{1}{2} \left[x - \frac{\sin 6x}{6} \right] + C$$

$$= \frac{x}{2} - \frac{\sin 6x}{12} + C \quad (D)$$

$$9. \int_0^k \frac{dx}{1+(x-1)^2} = \frac{\pi}{2}$$

$$\left[\tan^{-1}(x-1) \right]_0^k = \frac{\pi}{2}$$

$$\tan^{-1}(k-1) - \tan^{-1}(-1) = \frac{\pi}{2}$$

$$\tan^{-1}(k-1) + \frac{\pi}{4} = \frac{\pi}{2}$$

$$k-1 = 1$$

$$k = 2 \quad (C)$$

$$(10) p(x) = 2x^4 - 4x^3 - 10x^2 + 12x$$

$$p(3) = 2 \times 81 - 4 \times 27$$

$$-10 \times 9 + 36$$

$$= 0$$

$$\therefore (x-3) \text{ is a factor}$$

(C)

Question 11 (12 marks)

page 2

$$1a) \frac{6}{5x-2} \leq 2$$

Multiply by $(5x-2)^2$, $x \neq \frac{2}{5}$

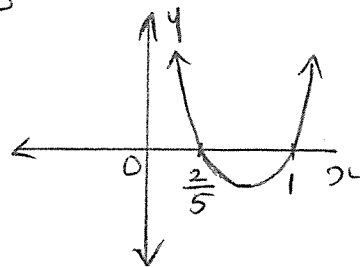
$$6(5x-2) \leq 2(5x-2)^2$$

$$(5x-2)^2 - 3(5x-2) \geq 0$$

$$(5x-2)(5x-5) \geq 0$$

x intercepts

$$x = \frac{2}{5} = 0.4, x = 1$$



From the graph

$$(5x-2)(5x-5) \geq 0$$

$$\text{When } x \leq \frac{2}{5} \text{ or } x \geq 1$$

$$\text{But } x \neq \frac{2}{5}$$

$$\therefore \underline{\underline{x < \frac{2}{5} \text{ or } x \geq 1}}$$

(3)

$$(b) \text{ LHS} = \cot 2x + \cot x$$

$$= \frac{\cos 2x}{\sin 2x} + \frac{\cos x}{\sin x}$$

$$= \frac{\cos 2x \sin x + \cos x \sin 2x}{\sin 2x \sin x}$$

(2)

$$= \frac{\sin 3x}{\sin 2x \sin x} = \text{RHS}$$

$$(c) u = \ln 3x, \frac{du}{dx} = \frac{1}{x}$$

$$du = \frac{1}{x} dx$$

$$\int \frac{du}{u^2} = \int u^{-2} du = -\frac{1}{u} + C$$

(3)

$$= \underline{\underline{-\frac{1}{\ln 3x} + C}}$$

(d) (i) $\cos x$ is undefined

when $1 - x^2 \leq 0$

$$(1+x)(1-x) \leq 0$$

$$x \leq -1 \text{ or } x \geq 1$$

(ii) $x = \sin u$

$$\frac{dx}{du} = \cos u$$

when $x=0$, $0 = \sin u$

$$u = \sin^{-1}(0) = 0$$

when $x = \frac{1}{2}$, $\frac{1}{2} = \sin u$

$$u = \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$$

$$\frac{\pi}{6} \int \frac{2 \sin u \cos u}{\cos u} du$$

$$= 2 \int_0^{\frac{\pi}{6}} \sin u \, du$$

$$= -2 \left[\cos u \right]_0^{\frac{\pi}{6}}$$

$$= -2 \left(\cos \frac{\pi}{6} - \cos 0 \right)$$

$$= -2 \left(\frac{\sqrt{3}}{2} - 1 \right) \quad (3)$$

$$= \underline{\underline{2 - \sqrt{3}}}$$

page 3

Question 12 (12 marks)

(a) $5 \sin x + 12 \cos x = A \sin(x + \alpha)$

$$= A \sin x \cos \alpha + A \cos x \sin \alpha$$

$$A \cos \alpha = 5$$

$$A \sin \alpha = 12$$

$$A^2 \sin^2 \alpha + A^2 \cos^2 \alpha = 169$$

$$A^2 = 169$$

$$A = \sqrt{169}$$

$$= 13 \quad (\because A > 0)$$

(3)

$$\sin \alpha = \frac{12}{13}, \quad \cos \alpha = \frac{5}{13}$$

$$\alpha = 1.18^\circ$$

$$5 \sin x + 12 \cos x = 13 \sin(x + 1.18)$$

(ii) $13 \sin(x + 1.18) = 8$

$$\sin(x + 1.18) = \frac{8}{13} \quad 0 \leq x \leq \pi$$

$$u = x + 1.18$$

$$\sin u = \frac{8}{13} \quad 1.18 \leq u \leq 4.32$$

$$u = 0.66, \pi - 0.66$$

$$= 0.66, 2.48, 0.66 + 2\pi, 2.48 + 2\pi$$

$$= 0.66, 2.48, 6.94, 8.76$$

$$x + 1.18 = 2.48$$

$$\underline{\underline{x = 1.3}}$$

(2)

$$(b)(i) 5! = 120 \quad (1)$$

$$(ii) 4! \times 2 = 48 \quad (1)$$

$$(iii) \frac{120-48}{120} = \frac{72}{120} = \frac{3}{5} \quad (1)$$

$$(c) (1^2+1) \times 1! + (2^2+1) \times 2! + (3^2+1) \times 3! + \dots + (n^2+1)n! = n(n+1)!$$

$$n=1$$

$$LHS = (1^2+1) \times 1! = 2$$

$$LHS = RHS$$

$\therefore$ true for $n=1$

$$RHS = 1(1+1)! = 2$$

Assume true for $n=k$

$$(1^2+1) \times 1! + (2^2+1) \times 2! + \dots + (k^2+1) \times k! = k(k+1)! \quad \text{--- (1)}$$

To prove true for $n=k+1$

$$\begin{aligned} (1^2+1) \times 1! + (2^2+1) \times 2! + \dots + (k^2+1) \times k! + ((k+1)^2+1) \times (k+1)! \\ = (k+1)(k+1)! \\ = (k+1)(k+2)! \quad \text{--- (2)} \end{aligned}$$

LHS of (2)

$$\begin{aligned} &= (1^2+1) \times 1! + (2^2+1) \times 2! + \dots + (k^2+1) \times k! + ((k+1)^2+1) \times (k+1)! \\ &= k(k+1)! + ((k+1)^2+1) \times (k+1)! \quad (\text{by assumption (1)}) \\ &= (k+1)! [k + (k+1)^2 + 1] \\ &= (k+1)! (k^2 + 3k + 2) \\ &= (k+1)! (k+1)(k+2) \\ &= (k+1)(k+2)! = RHS \text{ of (2)} \end{aligned}$$

4 marks

If the result is true for $n=k$, then it is true for $n=k+1$.

Hence by the principle of mathematical induction, the result is true for all positive integers $n \geq 1$.

Question 13 (12 marks)

(a)(i) $y = \tan^{-1}(x-1)$

$$\tan y = x-1$$

$$x = 1 + \tan y$$

$$V = \pi \int_0^{\frac{\pi}{4}} x^2 dy \quad (1)$$

$$= \pi \int_0^{\frac{\pi}{4}} (1 + \tan y)^2 dy$$

(ii) $V = \pi \int_0^{\frac{\pi}{4}} (1 + 2 \tan y + \tan^2 y) dy$

$$= \pi \int_0^{\frac{\pi}{4}} \left(2 \frac{\sin y}{\cos y} + \sec^2 y \right) dy$$

$$= \pi \left[-2 \log_e(\cos y) + \tan y \right]_0^{\frac{\pi}{4}}$$

$$= \pi \left\{ \left(-2 \log_e\left(\frac{1}{\sqrt{2}}\right) + 1 \right) - \left(-2 \log_e(1) + 0 \right) \right\}$$

$$= \pi \left(-2 \log_e\left(\frac{1}{\sqrt{2}}\right) + 1 \right)$$

$$= \pi \left(\log_e(2^{-\frac{1}{2}})^{-2} + 1 \right)$$

$$= \pi \left(\log_e 2 + 1 \right) \quad (3)$$

(b) $x = Vt \cos \theta \quad (1)$

$$y = Vt \sin \theta - \frac{1}{2} g t^2 \quad (2)$$

From (1) $t = \frac{x}{V \cos \theta}$

Substitute in (2)

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$$y = V \times \frac{x}{V \cos \theta} \times \sin \theta - \frac{1}{2} g \left(\frac{x}{V \cos \theta} \right)^2$$

$$= x \tan \theta - \frac{1}{2} \frac{g x^2}{V^2 \cos^2 \theta}$$

$$= x \tan \theta - \frac{g x^2}{2 V^2} \sec^2 \theta \quad (2)$$

$$= x \tan \theta - \frac{g x^2}{2 V^2} (1 + \tan^2 \theta) \quad (3)$$

(c) substitute $x = 240$, $y = 80$,
 $g = 10$ and $V = 60$ in (3)

$$80 = 240 \tan \theta - \frac{10 \times 240^2}{2 \times 60^2} (1 + \tan^2 \theta)$$

$$80 = 240 \tan \theta - 80(1 + \tan^2 \theta)$$

$$1 = 3 \tan \theta - (1 + \tan^2 \theta)$$

$$3 \tan \theta - (1 + \tan^2 \theta) = 1$$

$$\tan^2 \theta - 3 \tan \theta + 2 = 0$$

$$(\tan \theta - 1)(\tan \theta - 2) = 0$$

$$\tan \theta = 1 \text{ or } \tan \theta = 2$$

Since $\beta < \alpha$

$$\tan \beta = 1 \text{ and } \tan \alpha = 2 \quad (3)$$

$$(ii) xL = vt \cos \alpha$$

$$240 = 60(t+T) \cos \alpha$$

$$240 = 60t \cos \beta$$

$$4 = (t+T) \cos \alpha$$

$$4 = t \cos \beta$$

$$\frac{4}{\cos \alpha} = t+T$$

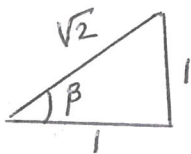
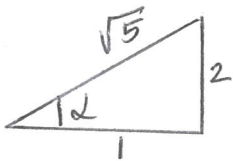
$$\frac{4}{\cos \beta} = t \quad (3)$$

$$t+T = 4 \sec \alpha$$

$$t = 4 \sec \beta$$

$$T = 4 \sec \alpha - 4 \sec \beta$$

$$= 4(\sec \alpha - \sec \beta)$$



$$\cos \alpha = \frac{1}{\sqrt{5}} \quad \cos \beta = \frac{1}{\sqrt{2}}$$

$$\sec \alpha = \sqrt{5} \quad \sec \beta = \sqrt{2}$$

$$\underline{T = 4(\sqrt{5} - \sqrt{2}) \text{ seconds}}$$

Question 14 (12 marks) Page 6

$$(a)(i) y = xL \cos^{-1} xL - \sqrt{1-xL^2}$$

$$\frac{dy}{dx} = xL \frac{x-1}{\sqrt{1-xL^2}} + \cos^{-1} xL - \frac{1}{2\sqrt{1-xL^2}} x^{-2xL} \quad (2)$$

$$= \frac{-xL}{\sqrt{1-xL^2}} + \cos^{-1} xL + \frac{xL}{\sqrt{1-xL^2}} = \cos^{-1} xL$$

$$(ii) \int_0^{\frac{1}{2}} \cos^{-1} xL dxL = \left[xL \cos^{-1} xL - \sqrt{1-xL^2} \right]_0^{\frac{1}{2}}$$

$$= \left(\frac{1}{2} \cos^{-1} \frac{1}{2} - \sqrt{1-\frac{1}{4}} \right) - (0 - \sqrt{1})$$

$$= \frac{1}{2} \times \frac{\pi}{3} - \sqrt{\frac{3}{4}} + 1 \quad (2)$$

$$= \frac{\pi}{6} - \frac{\sqrt{3}}{2} + 1$$

$$(b) p(x) = x^4 - 5x^3 - 9x^2 + 81x - 108$$

$$p'(x) = 4x^3 - 15x^2 - 18x + 81$$

$$p''(x) = 12x^2 - 30x - 18$$

$$p''(x) = 0$$

$$12x^2 - 30x - 18 = 0$$

$$2x^2 - 5x - 3 = 0 \quad (3)$$

$$(x-3)(2x+1) = 0$$

$$x = 3 \quad \text{or} \quad x = -\frac{1}{2}$$

$$p(3) = 81 - 135 - 81 + 243 - 108 = 0$$

$x = 3$ is the triple root.

Let the roots be 3, 3, 3 and α

$$\alpha + 9 = 5; \quad \alpha = -4$$

The roots are 3, 3, 3, -4

$$(c)(i) T = 25 + Ae^{-kt}$$

$$\text{LHS} = \frac{dT}{dt} = Ae^{-kt} \times -k \\ = -kAe^{-kt}$$

$$\text{RHS} = -k(T - 25) \\ = -k \times Ae^{-kt} \\ = -kAe^{-kt} \quad (2)$$

$$\text{LHS} = \text{RHS}$$

$\therefore T = 25 + Ae^{-kt}$ is a solution of $\frac{dT}{dt} = -k(T - 25)$

(ii) when $t = 10$, $T = 16^\circ\text{C}$

$$T(t) = 25 + Ae^{-kt}$$

$$T(0) = 10 = 25 + A$$

$$A = 10 - 25 = -15$$

$$16 = 25 - 15e^{-10k}$$

$$e^{-10k} = \frac{9}{15} \quad (3)$$

when $t = 40$, $T = ?$

$$T = 25 - 15e^{-40k}$$

$$= 25 - 15e^{4(-10k)}$$

$$= 25 - 15 \times \left(\frac{9}{15}\right)^4$$

$$= \underline{\underline{23.056}}$$

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Question 15 (12 marks)

$$(a) m(10m - 17) + 6 = 0$$

$$10m^2 - 17m + 6 = 0$$

$$10m^2 - 5b - 12b + 6 = 0$$

$$5b(2b - 1) - 6(2b - 1) = 0 \quad (3)$$

$$(2b - 1)(5b - 6) = 0$$

$$b = \frac{1}{2} \quad \text{or} \quad b = \frac{6}{5}$$

$$(b) \underline{u} \cdot \underline{w} = 2b + 2b = 4b$$

$$|\underline{u}| |\underline{w}| \cos \theta = \sqrt{b^2 + 4} \cdot \sqrt{b^2 + 4} \cos 60^\circ \\ = (b^2 + 4) \times \frac{1}{2}$$

$$4b = (b^2 + 4) \times \frac{1}{2}$$

$$8b = b^2 + 4$$

$$b^2 - 8b + 4 = 0$$

$$b = \frac{8 \pm \sqrt{64 - 4 \times 1 \times 4}}{2}$$

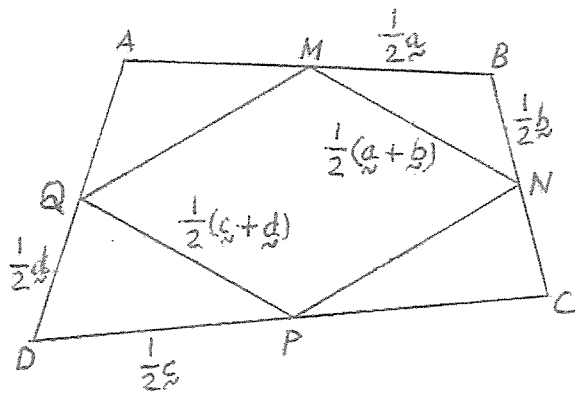
$$= \frac{8 \pm \sqrt{48}}{2}$$

$$= \frac{8 \pm 4\sqrt{3}}{2} \quad (4)$$

$$= \frac{4(2 \pm \sqrt{3})}{2}$$

$$= \underline{\underline{2(2 \pm \sqrt{3})}}$$

(C) (i)



$$\begin{aligned}
 \vec{a} + \vec{b} + \vec{c} + \vec{d} &= \vec{AB} + \vec{BC} + \vec{CD} + \vec{DA} \\
 &= \vec{AC} + \vec{CB} + \vec{DA} \\
 &= \vec{AB} + \vec{DA} = \vec{0}
 \end{aligned}$$

(2)

(ii) $\vec{MN} = \vec{MB} + \vec{BN}$

$$= \frac{1}{2} \vec{a} + \frac{1}{2} \vec{b}$$

$$= \frac{1}{2} (\vec{a} + \vec{b})$$

$$\vec{PQ} = \vec{PB} + \vec{BQ}$$

$$= \frac{1}{2} \vec{c} + \frac{1}{2} \vec{d}$$

$$= \frac{1}{2} (\vec{c} + \vec{d})$$

(3)

But $\vec{a} + \vec{b} + \vec{c} + \vec{d} = \vec{0}$

$$\vec{a} + \vec{b} = -(\vec{c} + \vec{d})$$

$$\therefore \vec{MN} = \vec{PQ} \text{ and } MN \parallel PQ$$

$\therefore$ MNPQ is a parallelogram.